WSClock Page Replacement Algorithm

1 Core Idea

Definition

WSClock combines the circular scanning idea of the **clock algorithm** with working-set information such as time of last use, R bit, and M bit.

The basic working set algorithm may scan the whole page table at every page fault. WSClock improves this by keeping only resident page frames in a circular list and scanning from a clock hand.

Main Goal

Find a page that is old enough to be outside the working set and cheap enough to evict.

2 Data Structure

WSClock keeps page frames in a circular list. Each entry stores:

Field	Meaning
Time of last use	Approximate virtual time when this page was last referenced.
R **bit**	Set when the page was referenced during the current clock interval.
M **bit**	Set when the page was modified, meaning it is dirty.
Clock hand	Points to the next candidate page frame to inspect.

Age Formula

$$\boxed{\text{age} = \text{current virtual time} - \text{time of last use}}$$

3 WSClock Figure

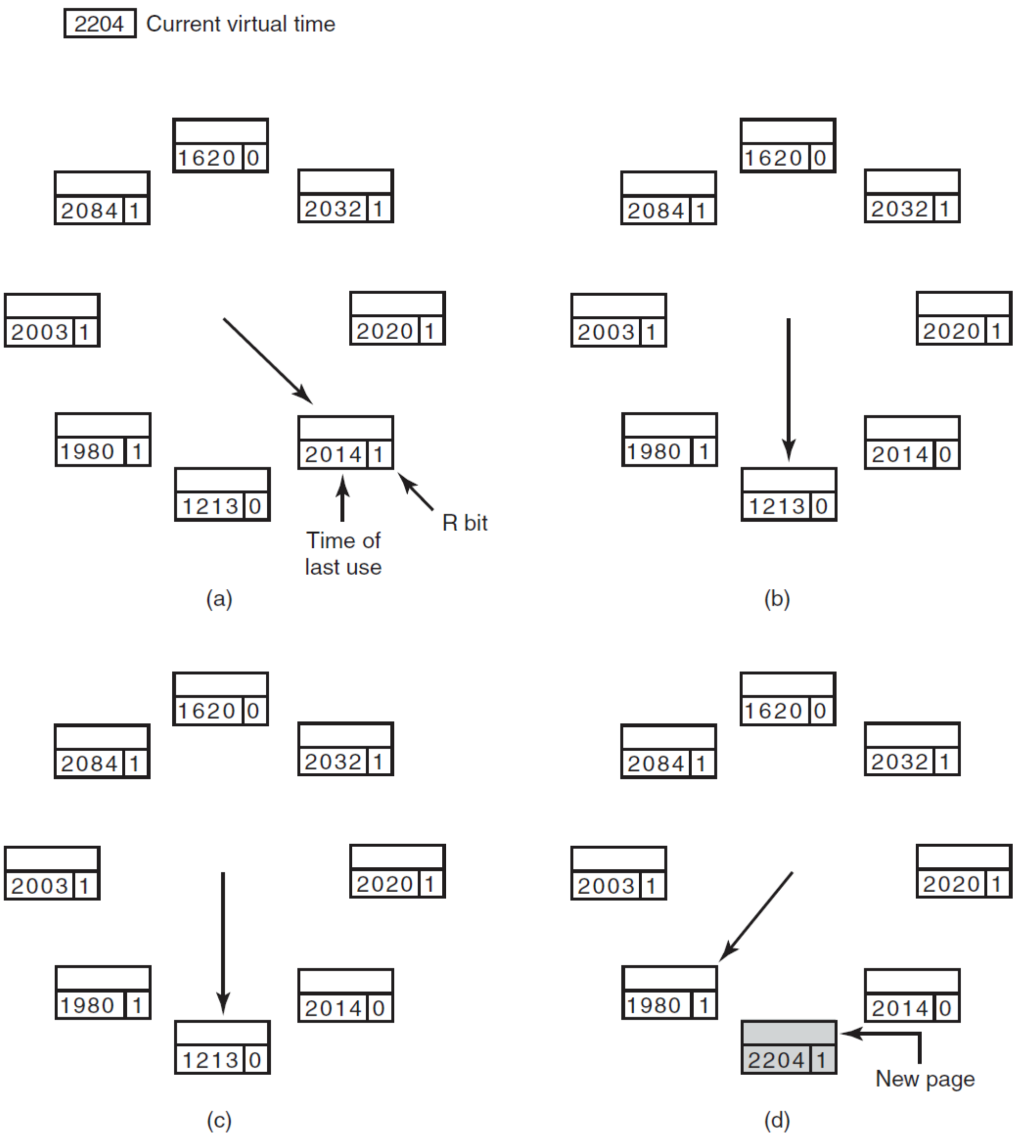

Figure 3-20. Operation of the WSClock algorithm. (a) and (b) give an example of what happens when $R = 1$. (c) and (d) give an example of $R = 0$.

Figure Setup

The current virtual time is 2204. Each small box shows a page frame: the larger number is the time of last use, and the small right-side number is the R bit.

4 Basic Scan Rule

At a page fault, WSClock first examines the page under the hand.

Observed page	Action
$R = 1$	Page was referenced recently. Clear R, update time of last use to current virtual time, advance the hand.
$R = 0$, **age** $> \tau$, **clean**	Page is outside the working set and has a valid disk copy. Evict it immediately.
$R = 0$, **age** $> \tau$, **dirty**	Schedule disk write, advance the hand, and keep searching.
$R = 0$, **age** $\leq \tau$	Page is still in the working set. Keep it and advance the hand.

5 Case 1: $R = 1$

If the hand points to a page with $R = 1$, the page was used during the current clock tick.

1. The page is treated as active.
2. Its R bit is cleared to 0.
3. Its time of last use is updated to the current virtual time.
4. The hand advances to the next page.

Panels (a) and (b)

In the figure, panels (a) and (b) show this case. The hand finds a page with $R = 1$, clears its R bit, and moves forward.

6 Case 2: $R = 0$

If $R = 0$, the page was not referenced during the current clock tick. Now WSClock checks its age.

$$\text{age} = 2204 - \text{time of last use}$$

For the page with time of last use 1213:

$$\text{age} = 2204 - 1213 = 991$$

If τ is smaller than 991, then this page is outside the working set.

7 Clean Old Page

Best Victim

If $R = 0$, age $> \tau$, and $M = 0$, the page is old and clean. WSClock evicts it immediately.

Clean means the disk copy is already up to date. The incoming page can simply replace the old page in that frame.

Panels (c) and (d)

Panels (c) and (d) show this case. The old page with time 1213 and $R = 0$ is replaced by the new page. The new page receives time 2204.

8 Dirty Old Page

If a page is outside the working set but dirty, it cannot be reused immediately.

Condition	Action
$R = 0$, **age** $> \tau$, $M = 1$	Schedule a disk write for the page.
After scheduling write	Advance the hand and continue searching for another victim.
Reason	A clean old page may appear soon and can be used immediately.

Disk Traffic Control

In practice, the OS may limit how many dirty pages are scheduled for write-back during one sweep to avoid excessive disk traffic.

9 If the Hand Makes a Full Circle

Sometimes the hand returns to its starting point without finding an immediate clean victim.

Full-circle case	What WSClock does
At least one write was scheduled	Keep scanning. Eventually a scheduled write completes, making some page clean. Evict the first clean old page found.
No writes were scheduled	All pages appear to be in the working set. Use a clean page if one was seen; otherwise choose the current page and write it back if needed.

10 Step-by-Step Algorithm

1. Start at the page frame under the clock hand.
2. If $R = 1$, clear R, update last-use time, advance hand.
3. If $R = 0$, compute age.
4. If age $\leq \tau$, keep the page and advance hand.
5. If age $> \tau$ and $M = 0$, evict the page.
6. If age $> \tau$ and $M = 1$, schedule write-back and advance hand.
7. Continue until a usable victim is found.

11 Why WSClock Is Practical

Advantage	Explanation
Avoids full page-table scan	It scans resident frames using a circular list.
Uses recency information	R and last-use time estimate the working set.
Handles dirty pages sensibly	Dirty old pages are scheduled for write-back instead of blocking immediately.
Good practical behavior	It is simpler than exact working set replacement and performs well.

12 Final Summary

One-Box Summary

WSClock = Clock algorithm + working-set information.
If $R = 1$, clear R and advance the hand.
If $R = 0$ and age $> \tau$ and clean, evict.
If dirty, schedule write-back and keep searching.